FHV
Vorarlberg University
of Applied Sciences

Interference
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EIT & MEC Bachelor

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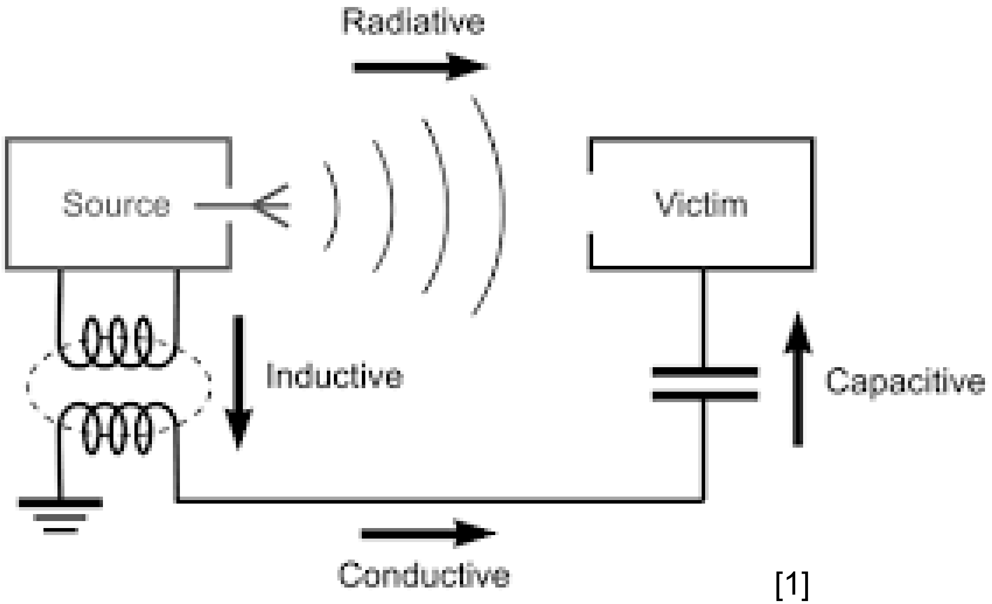
Coupling Mechanisms

Four different mechanisms need to be considered:

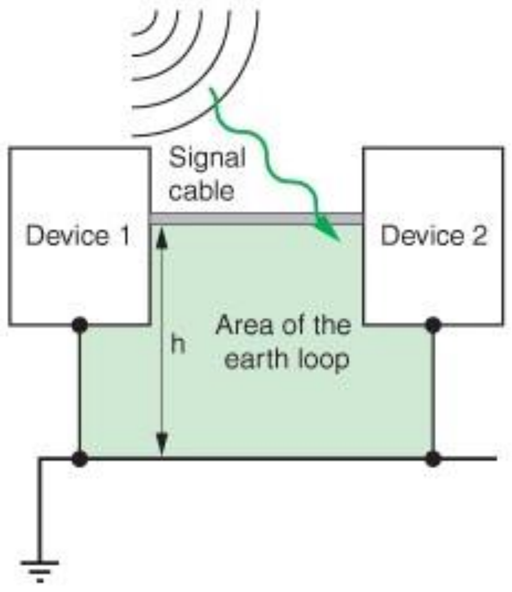
- inductive coupling
- coupling by radiation
- galvanic coupling
- capacitive coupling



[1]

Coupling by Radiation

- Free electromagnetic waves



Example of field-to-loop coupling

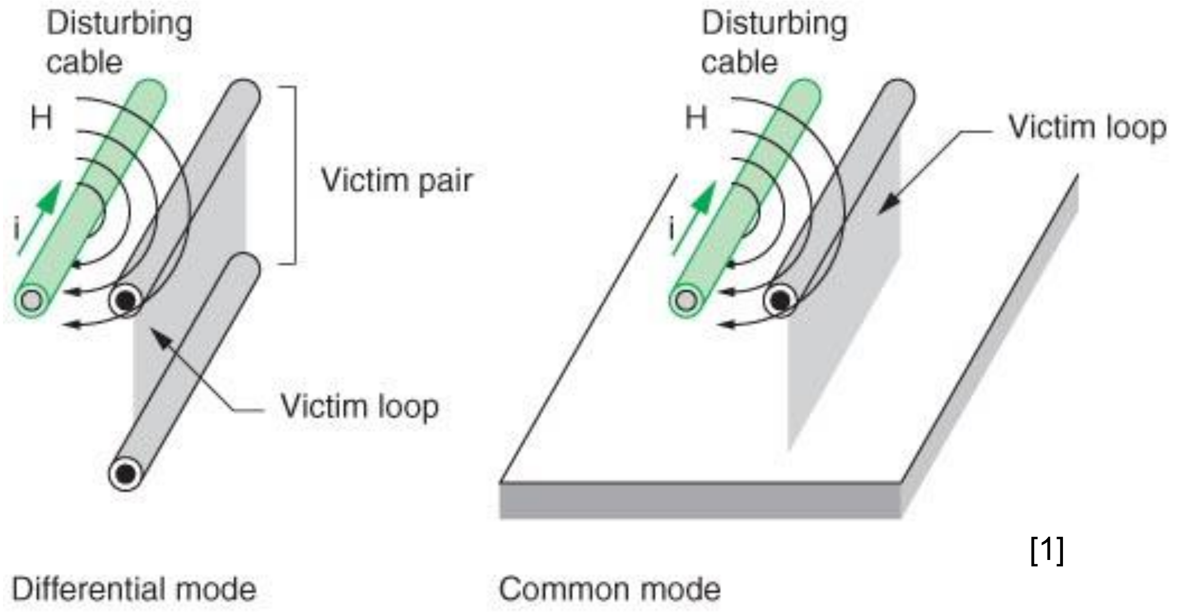
[1]

What to do:

- Reduce loop area
- Shielding with copper/aluminum

Inductive Coupling I

- Coupling by magnetic fields
- Given by dI/dt – increases with frequency
- Examples: short-circuit, fault-current, lightning-strike



[1]

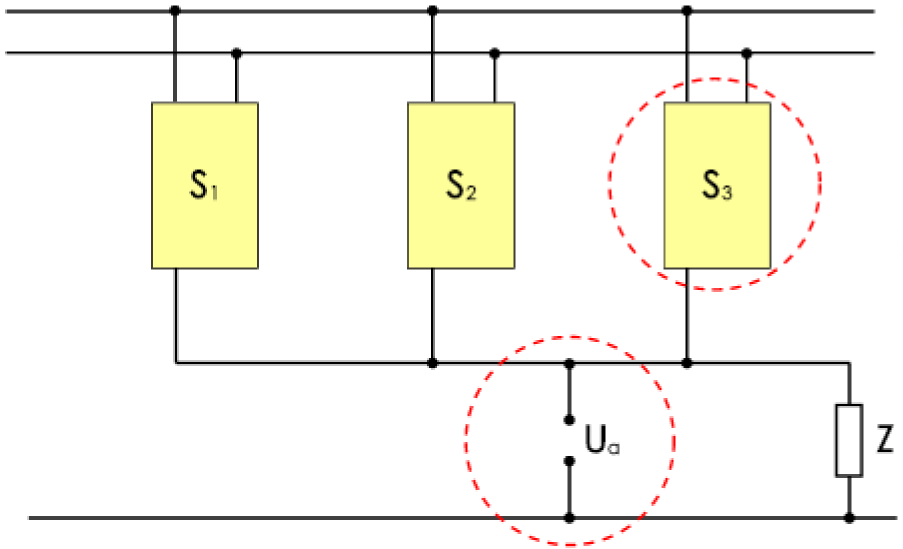
Inductive Coupling II

What to do:

- Reduce loop-area (e.g. twisted wires)
- Reduce length of wires/tracks
- Keep dI/dt as small as possible
- Change orientation / position
- Shielding with high-permeable material for low frequency and copper/aluminum for f > 100Hz – 1kHz

Galvanic Coupling I

- Common currents through wires and tracks
- Influence between to circuits by voltage drop
- coupling even at low frequencies
- Supply-wires or common ground-wires introduce distortions from one module to another



[1]

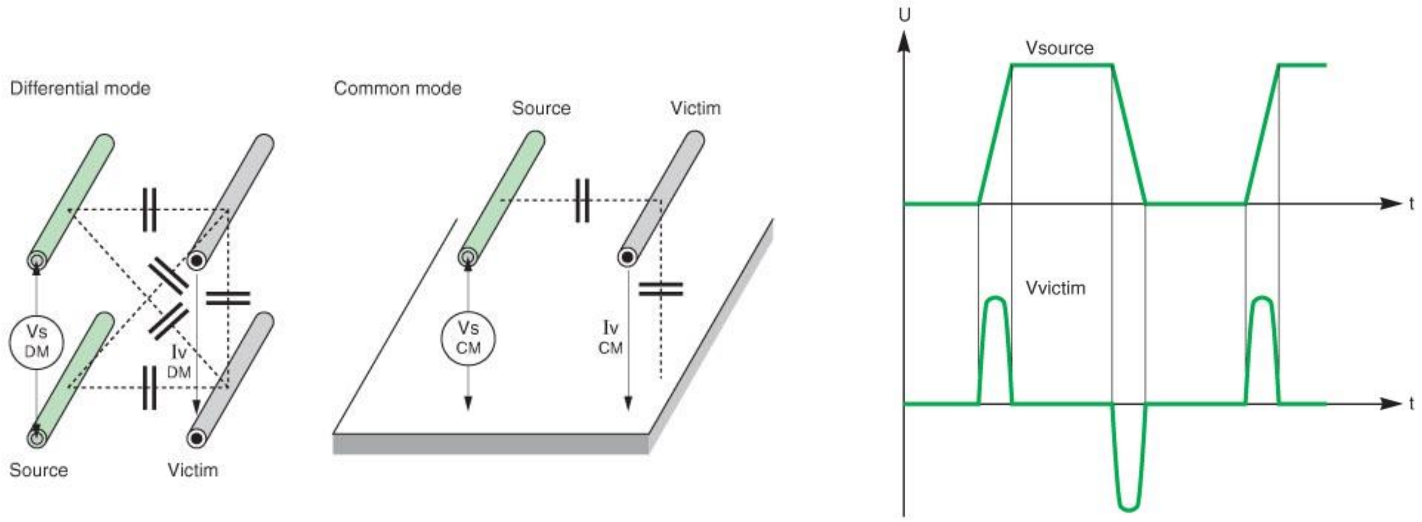
Galvanic Coupling II

What to do:

- Separate supply-lines for different modules
- Bypass capacitors reduce at least peak-currents
- “Star point”-grounding
- Low inductive grounding (short)
- Common-mode chokes

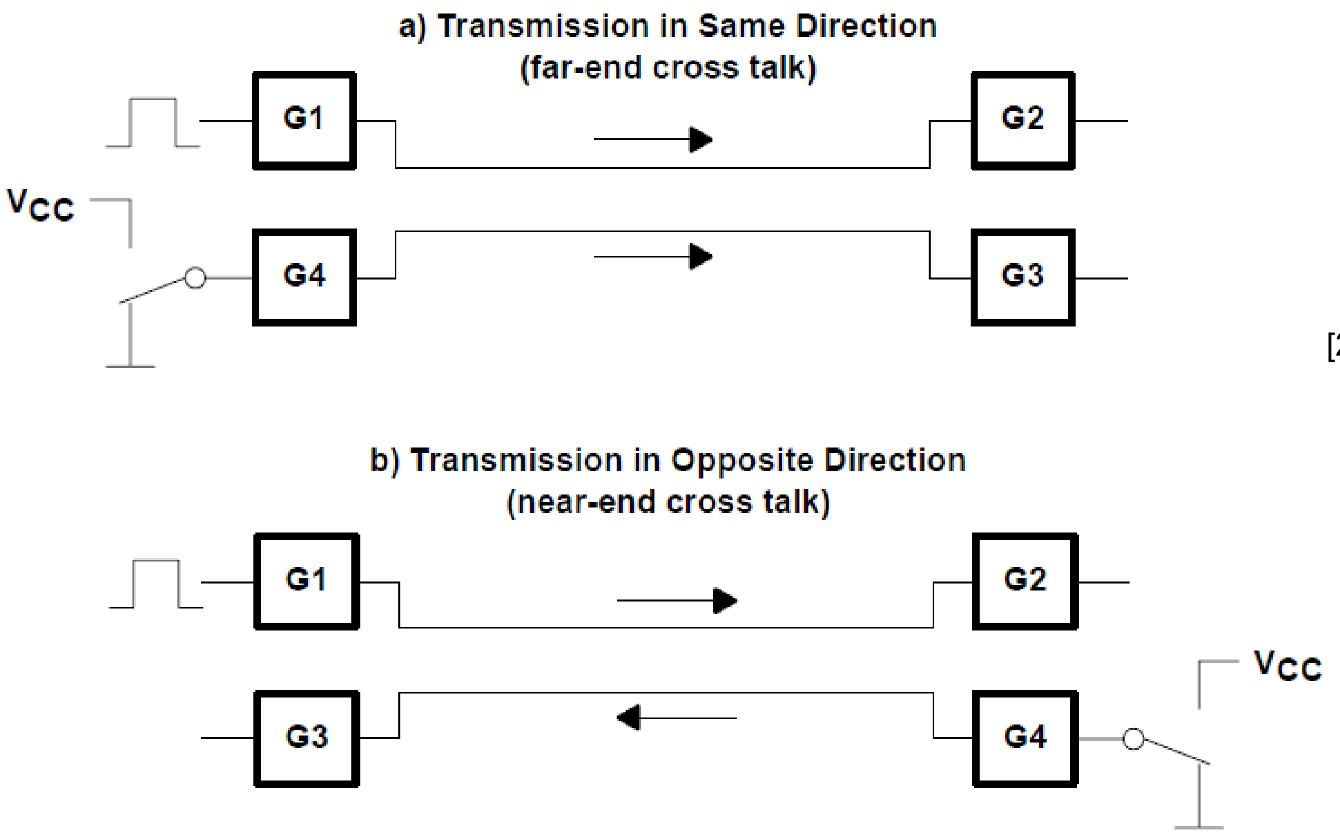
Capacitive Coupling (Crosstalk) I

- Coupling by electric fields
- Given by dv/dt – increases with frequency



[1]

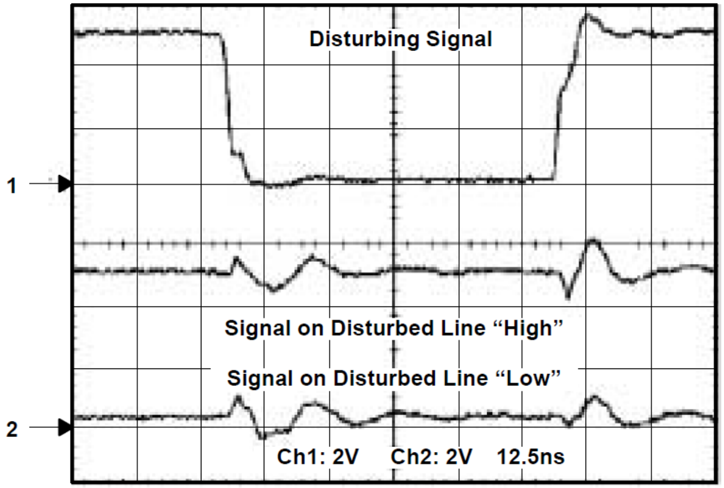
Capacitive Coupling (Crosstalk) II



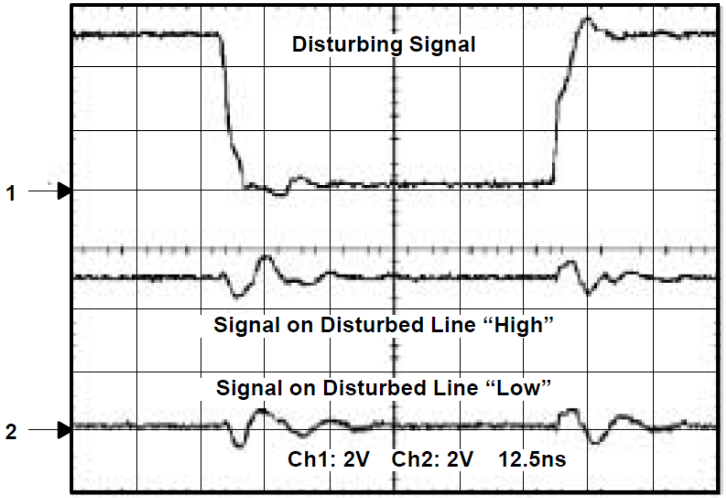
[2], p. 24

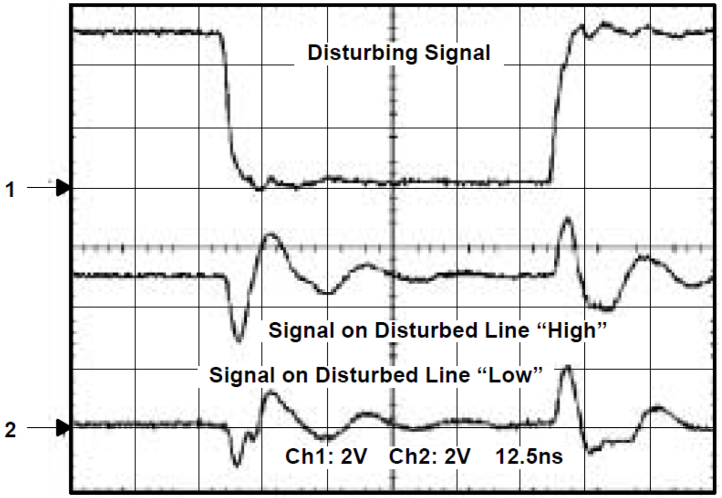
Capacitive Coupling (Crosstalk) III

- Far end (top right)
- Near end (bottom right)
- Near end with screening (bottom left)



[2], p. 24 - 26





Capacitive Coupling (Crosstalk) IV

What to do:

- No parallel wiring and reduced length of wires/tracks
- Use “guard-tracks” or ground-wires
- Shielding with copper/aluminum
- Increase distance between circuits
- Use Low ohmic circuits
- Keep dV/dt as small as possible

References

[1] Lecture slides in digital and microprocessor engineering by Martin Greif, 2017

[2] AHC/AHCT Designer’s Guide,Texas Instruments Corporation, Sept. 1998, Document SCLA013D. http://focus.ti.com/lit/ml/scla013d/scla013d.pdf

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